MULTIPLICATION BY A POWER OF F IDENTIFIES THE COHOMOLOGY OF M UP TO F-POWER TORSION WITH THE COHOMOLOGY OF ETA_F M

THE STACKS PROJECT AUTHORS

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Editorial note. The notation eta_f M of the statement and proof denotes the shifted complex with i-th term f^i M^i, and it is written by the author with the ordinary Greek letter eta and the subscript f; no macro beyond the shared preamble is needed, and the example that precedes the lemma at source lines 27480-27491 is outside the counted spans.

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Allowed standard prerequisites (editorial).

- The shifted complex of a complex M with respect to a nonzerodivisor f has i-th term f^i M^i and differentials induced by the differentials of M. used at proof lines 27504-27506, where the cycles of the shifted complex are computed. Turns the cohomology of the shifted complex into a quotient of f^i times the cycles of M.
- The kernel of the differential in degree i of the shifted complex equals f^i ker(d^i), because multiplication by f^i is injective on M^i and on M^{i+1} when f is a nonzerodivisor on all terms. used at proof lines 27505-27506. Shows that multiplication by f^i lands in the cycles of the shifted complex and induces a surjection on cohomology.
- An element of the kernel of the induced map on cohomology satisfies f^i z = d(f^{i-1} y), which after cancelling f^i and f^{i-1} means f z = d(y), that is z is f-torsion in cohomology. used at proof lines 27510-27514. Identifies the kernel of the surjection with the f-torsion subgroup of H^i(M), which yields the displayed isomorphism.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let A be a ring and let $f \in A$ be a nonzerodivisor. Let $M^\bullet$ be a complex of f-torsion free A-modules. There is a canonical isomorphism*

$$f^i : H^i(M^\bullet)/H^i(M^\bullet)[f] \longrightarrow H^i(\eta_f M^\bullet)$$

given by multiplication by f^i.

Proof. Observe that $\mathrm{Ker}(d^i : (\eta_f M)^i \to (\eta_f M)^{i+1})$ is equal to $\mathrm{Ker}(d^i : f^i M^i \to f^i M^{i+1}) = f^i \mathrm{Ker}(d^i : M^i \to M^{i+1})$. This we get a surjection $f^i : H^i(M^\bullet) \to H^i(\eta_f M^\bullet)$ by sending the class of $z \in \mathrm{Ker}(d^i : M^i \to M^{i+1})$ to the class of $f^i z$. If we obtain the zero class in $H^i(\eta_f M^\bullet)$ then we see that $f^i z = d^{i-1}(f^{i-1} y)$ for some $y \in M^{i-1}$. Since f is a nonzerodivisor on all the modules involved, this means $fz = d^{i-1}(y)$ which exactly means that the class of z is f-torsion as desired. $\square$